



**VIDYA VIKAS INSTITUTE OF  
ENGINEERING AND TECHNOLOGY,  
MYSORE**

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**VTU Previous Question Papers**

**Civil Engineering**

**III, V & VI Semester**

**Dec/Jan-2024**

***VVIET***

***Bannur road, Alanahalli, Mysore-570028***

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# CBCS SCHEME

USN

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BCV301

## Third Semester B.E/B.Tech. Degree Examination, Dec.2023/Jan.2024 Strength of Materials

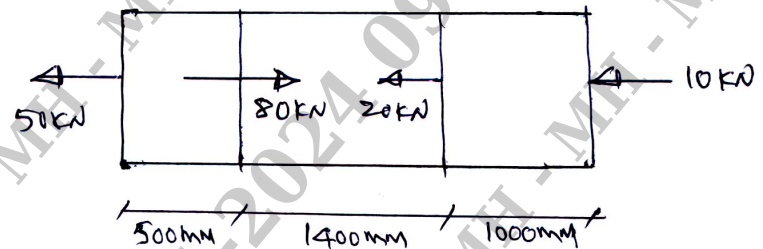
Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**

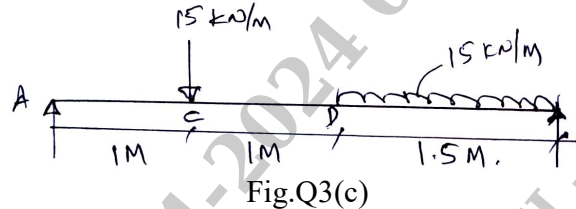
**2. M : Marks , L: Bloom's level , C: Course outcomes.**

**3. Missing data, if any, may be suitably assumed.**

|           |    | Module – 1                                                                                                                                                                                                                                                                                                                                                                                                                 | M  | L  | C   |
|-----------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 1         | a. | Define the following terms :<br>i) Poisson's ratio<br>ii) Modulus of rigidity<br>iii) Impact load<br>iv) Volumetric strain.                                                                                                                                                                                                                                                                                                | 4  | L1 | CO1 |
|           | b. | The following data refers to mild steel tested on a lab. Diameter of the specimen 25mm, length of the specimen 300mm, Extension under a load of 15kN is 0.045mm, load at yield point 127.65kN, max load 208.60kN, length of the specimen at failure 375mm, diameter at failure 17.75mm. Determine young's modulus, yield strength, ultimate strength, % elongation of the specimen, % decrease in c/s at area of specimen. | 10 | L3 | CO1 |
|           | c. | A brass bar having cross-sectional area 300sq.mm is subjected to axial forces as shown in the Fig.Q1(c). Determine the total elongation of the bar taking $E = 84\text{GPa}$ .<br><br><div style="text-align: center;">  <p>Fig.Q1(c)</p> </div>                                                                                       | 6  | L3 | CO1 |
| <b>OR</b> |    |                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |     |
| 2         | a. | A steel rod 20mm diameter, length 6m is connected at the ends to a pair of walls at a temperature of $120^{\circ}\text{C}$ . Find the pull exerted on the wall if the temperature falls to $40^{\circ}\text{C}$ when :<br>i) Supports don't yield<br>ii) Supports yield by 1.1mm, take $E = 200\text{GPa}$ . $\alpha = 12 \times 10^{-6}/^{\circ}\text{C}$ .                                                               | 6  | L3 | CO1 |
|           | b. | Derive the relationship between modulus of elasticity, modulus of rigidity and Poisson's ratio.                                                                                                                                                                                                                                                                                                                            | 6  | L3 | CO1 |
|           | c. | A load of 270kN is acting on a short RCC column of size $(200 \times 200)\text{mm}^2$ . The column is reinforced with 10 bars of 12mm diameter. Determine the loads and the corresponding stresses on steel and concrete. Take $E_s = 16.5E_c$ .                                                                                                                                                                           | 8  | L3 | CO1 |

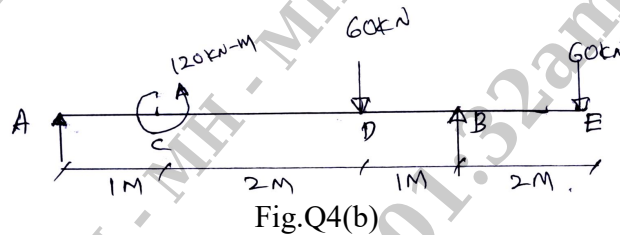
## Module – 2

|   |    |                                                                                                                                                                                                          |    |    |     |
|---|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 3 | a. | Define Hogging Bending moment and sagging bending moment.                                                                                                                                                | 4  | L2 | CO2 |
|   | b. | Derive the relationship between loading, shear force and bending moment.                                                                                                                                 | 6  | L3 | CO2 |
|   | c. | A simply supported beam is subjected to point load of 15kN together with a udl of 15kN/m as shown in the Fig.Q3(c). Draw SFD and BMD. Find also point of low shear and the corresponding bending moment. | 10 | L3 | CO2 |



## OR

|   |    |                                                                                                                               |    |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 4 | a. | Show that maximum bending moment for a simply supported beam carrying udl of intensity $w$ /unit length is $\frac{wl^2}{8}$ . | 6  | L2 | CO2 |
|   | b. | Draw SFD and BMD for an overhanging beam carrying forces as shown in the Fig.Q4(b).                                           | 14 | L3 | CO2 |

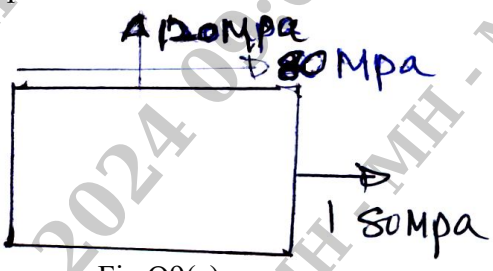


## Module – 3

|   |    |                                                                                                                                                                                                                                       |   |    |     |
|---|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 5 | a. | Define :<br>i) Modulus of rupture<br>ii) Section modulus<br>iii) Flexural rigidity.                                                                                                                                                   | 6 | L1 | CO3 |
|   | b. | Derive bending equation with usual notation.                                                                                                                                                                                          | 6 | L3 | CO3 |
|   | c. | A simply supported beam of span 5m has cross section 150mm × 250mm. If the permissible stress is 10N/mm <sup>2</sup> . Find :<br>i) Maximum intensity of UDL it can carry<br>ii) Maximum concentrated load P applied 2m from one end. | 8 | L3 | CO3 |

## OR

|   |    |                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 6 | a. | List the assumptions made pure torsion.                                                                                                                                                                                                                                                                                                                                                                     | 6  | L1 | CO3 |
|   | b. | Derive torque – equation with usual notation.                                                                                                                                                                                                                                                                                                                                                               | 4  | L3 | CO3 |
|   | c. | A solid shaft has to transmit 250KW power at 100rpm. If the shear stress not to exceed 75MPa what should be the diameter of the shaft. If this shaft is to be replaced by a hollow shaft whose internal diameter, is 0.6 times the external diameter determine the size of the shaft. Also determine the saving on the weight of the material. Assume max shear stress remain the same for both the shafts. | 10 | L3 | CO3 |

| Module – 4                                                                                            |    |                                                                                                                                                                                                                                                                                                                                                                     |    |    |     |
|-------------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 7                                                                                                     | a. | Define slope, deflection and curvature.                                                                                                                                                                                                                                                                                                                             | 6  | L1 | CO4 |
|                                                                                                       | b. | Derive moment–curvature equation.                                                                                                                                                                                                                                                                                                                                   | 6  | L3 | CO4 |
|                                                                                                       | c. | A girder of uniform section and constant depth is simply supported over a span of 3m. If the point load at the mid span is 30kN and $I_{XX} = 15.614 \times 10^{-6} \text{m}^4$ , calculate : i) Central deflection ii) The slopes at the ends if the beam. Take $E = 200 \text{GN/m}^2$ .                                                                          | 8  | L3 | CO4 |
| OR                                                                                                    |    |                                                                                                                                                                                                                                                                                                                                                                     |    |    |     |
| 8                                                                                                     | a. | Differentiate between long columns and short columns.                                                                                                                                                                                                                                                                                                               | 4  | L1 | CO4 |
|                                                                                                       | b. | Derive Euler's Buckling load for long columns whose ends are hinged.                                                                                                                                                                                                                                                                                                | 6  | L3 | CO4 |
|                                                                                                       | c. | A hollow tube 6m length of external diameter 16mm and thickness 10mm is subjected to minimum crippling load. Find Euler's load for this column when : i) Both ends fixed ii) One end fixed and other end hinged. Take $E = 200 \text{GPa}$ .                                                                                                                        | 10 | L3 | CO4 |
| Module – 5                                                                                            |    |                                                                                                                                                                                                                                                                                                                                                                     |    |    |     |
| 9                                                                                                     | a. | Derive principle planes and principle stresses.                                                                                                                                                                                                                                                                                                                     | 4  | L1 | CO5 |
|                                                                                                       | b. | Differentiate between thin cylinders and thick cylinders.                                                                                                                                                                                                                                                                                                           | 4  | L1 | CO5 |
|                                                                                                       | c. | The state of stress at a point on a strained material is 120Mpa and is as shown in the Fig.Q9(c). Determine :<br>i) The direction of principal planes<br>ii) The magnitude of principal stresses<br>iii) The magnitude of maximum shear stress and its directions.<br>Sketch the stresses and planes.                                                               | 12 | L3 | CO5 |
|  <p>Fig.Q9(c)</p> |    |                                                                                                                                                                                                                                                                                                                                                                     |    |    |     |
| OR                                                                                                    |    |                                                                                                                                                                                                                                                                                                                                                                     |    |    |     |
| 10                                                                                                    | a. | Derive Lamé's equation with usual notation.                                                                                                                                                                                                                                                                                                                         | 8  | L3 | CO5 |
|                                                                                                       | b. | A shell 3.25m long; 1m in diameter is subjected to internal fluid pressure of 1MPa, if the thickness of the shell is 10mm find :<br>i) Hoop-stress<br>ii) Longitudinal stress<br>iii) Maximum shear stress<br>iv) Change in diameter and length<br>v) Volumetric strain and hence measure in volume<br>Take $E = 2 \times 10^3 \text{MPa}$ ; $\frac{1}{m} = 0.30$ . | 12 | L3 | CO5 |

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**Module – 3**

|            |           |                                                                     |           |           |            |
|------------|-----------|---------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.5</b> | <b>a.</b> | Discuss in detail about characteristics of contours.                | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|            | <b>b.</b> | Explain the procedure of conducting the L/S and C/S by using level. | <b>10</b> | <b>L2</b> | <b>CO3</b> |

**OR**

|            |           |                                                           |           |           |            |
|------------|-----------|-----------------------------------------------------------|-----------|-----------|------------|
| <b>Q.6</b> | <b>a.</b> | Discuss in detail about contouring using level.           | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|            | <b>b.</b> | Explain how coordinates are measured using total station. | <b>10</b> | <b>L2</b> | <b>CO3</b> |

**Module – 4**

|            |           |                                                                                          |           |           |            |
|------------|-----------|------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.7</b> | <b>a.</b> | For applying Rankine's method, provide the procedure of setting out of horizontal curve. | <b>10</b> | <b>L3</b> | <b>CO4</b> |
|            | <b>b.</b> | Explain the procedure of setting out two room building by center line method.            | <b>10</b> | <b>L2</b> | <b>CO4</b> |

**OR**

|            |           |                                                                                                                                                                                                                                                                                                                                                                            |           |           |            |
|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.8</b> | <b>a.</b> | Explain how areas are measured by trapezoidal and Simpson's rule.                                                                                                                                                                                                                                                                                                          | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|            | <b>b.</b> | A railway embankment is 10m wide with side slope $1\frac{1}{2}$ to 1. Assuming the ground to be level in a direction transverse to the center line, calculate the volume contained in a length of 120m, the center heights at 20m intervals being in meters are 2.200, 3.700, 3.800, 4.000, 3.800, 2.800, 2.500. Calculate volume by trapezoidal rule and prismoidal rule. | <b>10</b> | <b>L3</b> | <b>CO4</b> |

**Module – 5**

|            |           |                                                                                              |           |           |            |
|------------|-----------|----------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.9</b> | <b>a.</b> | What is absolute and differential positioning with GPS? Explain about Gagan system in India. | <b>10</b> | <b>L2</b> | <b>CO5</b> |
|            | <b>b.</b> | What are the applications and advantages of surveying with drone?                            | <b>10</b> | <b>L3</b> | <b>CO5</b> |

**OR**

|             |           |                                                                                        |           |           |            |
|-------------|-----------|----------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.10</b> | <b>a.</b> | What are the applications and uses of remote sensing and GIS in engineering surveying. | <b>10</b> | <b>L3</b> | <b>CO5</b> |
|             | <b>b.</b> | Outline the process of drone surveying.                                                | <b>10</b> | <b>L2</b> | <b>CO5</b> |

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## Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Engineering Geology

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                           | M  | L  | C   |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Explain the role of Geology in the Civil Engineering.                                                                                                                                     | 08 | L2 | CO1 |
|            | b. | What is Plate tectonics? Explain different type of plate boundaries.                                                                                                                      | 06 | L2 | CO1 |
|            | c. | Explain causes, effects of earthquake.                                                                                                                                                    | 06 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                           |    |    |     |
| Q.2        | a. | Explain briefly the internal structure of the earth.                                                                                                                                      | 10 | L2 | CO1 |
|            | b. | Write short notes on:<br>i) Tsunami – causes and effect<br>ii) Land Slides – causes and remedial measures                                                                                 | 10 | L1 | CO1 |
| Module – 2 |    |                                                                                                                                                                                           |    |    |     |
| Q.3        | a. | What is a mineral? Describe the following physical properties of a mineral.<br>i) Quartz                  ii) Hematite                  iii) Gypsum.                                      | 12 | L2 | CO2 |
|            | b. | Enumerate the classification of Igneous Rocks.                                                                                                                                            | 08 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                           |    |    |     |
| Q.4        | a. | What is Metamorphism? Explain the different kinds of metamorphism with example.                                                                                                           | 08 | L2 | CO2 |
|            | b. | Describe the following rocks for its geological/physical and engineering properties and its suitability as building material.<br>i) Granite                  ii) Lime Stone               | 08 | L2 | CO2 |
|            | c. | Describe the character of good building stones.                                                                                                                                           | 04 | L2 | CO2 |
| Module – 3 |    |                                                                                                                                                                                           |    |    |     |
| Q.5        | a. | What is weathering? Explain the types of weathering.                                                                                                                                      | 08 | L2 | CO3 |
|            | b. | Distinguish between Black Cotton Soil and Laterite Soil.                                                                                                                                  | 06 | L2 | CO3 |
|            | c. | Explain the effect of weathering on Monumental rock.                                                                                                                                      | 06 | L2 | CO3 |
| OR         |    |                                                                                                                                                                                           |    |    |     |
| Q.6        | a. | Explain soil formation and its profile.                                                                                                                                                   | 10 | L2 | CO3 |
|            | b. | Explain the classification of soil based on their grain size.                                                                                                                             | 06 | L2 | CO3 |
|            | c. | Write a note on drifted soil.                                                                                                                                                             | 04 | L1 | CO3 |
| Module – 4 |    |                                                                                                                                                                                           |    |    |     |
| Q.7        | a. | A bed of shale is dipping maximum of 32° along S45°E. Determine the amount of its apparent dip along S80°E. (Solve by Graphical / Trigonometric method).                                  | 06 | L3 | CO4 |
|            | b. | What are folds? Explain briefly the different types of folds with neat sketches.                                                                                                          | 10 | L2 | CO4 |
|            | c. | On a horizontal tunnel, a bed of sandstone dips 30° eastward. Its true thickness is 200m. Determine its vertical thickness and width of the out crop in the tunnel. [Scale : 1cm = 100m]. | 04 | L3 | CO4 |

| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |     |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.8        | a. | Three test bore holes (P, Q, R) are sunk at 3 point of an equilateral triangle, whose sides are 480m each. P is west of Q and R is midpoint of PQ. Bore holes P, Q and R reach the upper surface of a rich coal seam at 100 m, 220 m and 260 m depth respectively. Determine<br>i) The altitude [Dip and Strike] of the coal seam<br>ii) Another borehole 'S' is sunk exactly at midpoint of QR. Determine at what depth the new bore hole reaches the same coal seam. | 10 | L3 | CO4 |
|            | b. | Explain the impact of faults on Dam and Tunnel project.                                                                                                                                                                                                                                                                                                                                                                                                                | 04 | L2 | CO4 |
|            | c. | What are Faults? Explain the parts of Faults.                                                                                                                                                                                                                                                                                                                                                                                                                          | 06 | L2 | CO4 |
| Module – 5 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |     |
| Q.9        | a. | Define Aquifer. Explain briefly the different types of Aquifer with a neat sketch.                                                                                                                                                                                                                                                                                                                                                                                     | 12 | L2 | CO5 |
|            | b. | Explain water bearing properties of Rocks.                                                                                                                                                                                                                                                                                                                                                                                                                             | 08 | L2 | CO5 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |     |
| Q.10       | a. | Explain the electric resistivity method for exploration of Ground Water.                                                                                                                                                                                                                                                                                                                                                                                               | 12 | L2 | CO5 |
|            | b. | Explain the factors affecting permeability of Rocks.                                                                                                                                                                                                                                                                                                                                                                                                                   | 08 | L2 | CO5 |

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**Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024**  
**Water Supply and Waste Water Engineering**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*  
*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |        |                                                                                                                                                                                                                                                                                                                                                                                                     |        | M      | L      | C    |      |      |            |        |        |        |        |        |    |    |     |
|------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|------|------|------|------------|--------|--------|--------|--------|--------|----|----|-----|
| Q.1        | a.     | Briefly explain : i) Factors affecting per capita demand of water.<br>ii) Factors affecting design period.<br>iii) Domestic water demand.                                                                                                                                                                                                                                                           | 12     | L2     | CO1    |      |      |      |            |        |        |        |        |        |    |    |     |
|            | b.     | Explain the need and importance of protected water supply to the community.                                                                                                                                                                                                                                                                                                                         | 8      | L2     | CO1    |      |      |      |            |        |        |        |        |        |    |    |     |
| OR         |        |                                                                                                                                                                                                                                                                                                                                                                                                     |        |        |        |      |      |      |            |        |        |        |        |        |    |    |     |
| Q.2        | a.     | The population of 5 decades from 1980 to 2020 are given in below table. Find out the population after 3 decades beyond the last known decades by using arithmetic increase method.<br><table><tr><td>Year</td><td>1980</td><td>1990</td><td>2000</td><td>2010</td><td>2020</td></tr><tr><td>Population</td><td>25,000</td><td>28,000</td><td>34,000</td><td>42,000</td><td>47,000</td></tr></table> | Year   | 1980   | 1990   | 2000 | 2010 | 2020 | Population | 25,000 | 28,000 | 34,000 | 42,000 | 47,000 | 12 | L2 | CO1 |
| Year       | 1980   | 1990                                                                                                                                                                                                                                                                                                                                                                                                | 2000   | 2010   | 2020   |      |      |      |            |        |        |        |        |        |    |    |     |
| Population | 25,000 | 28,000                                                                                                                                                                                                                                                                                                                                                                                              | 34,000 | 42,000 | 47,000 |      |      |      |            |        |        |        |        |        |    |    |     |
|            | b.     | List different methods of population forecasting. Explain briefly Arithmetical and Geometrical Increase method.                                                                                                                                                                                                                                                                                     | 8      | L2     | CO1    |      |      |      |            |        |        |        |        |        |    |    |     |
| Module – 2 |        |                                                                                                                                                                                                                                                                                                                                                                                                     |        |        |        |      |      |      |            |        |        |        |        |        |    |    |     |
| Q.3        | a.     | Explain the objectives of Water treatment or Water purification.                                                                                                                                                                                                                                                                                                                                    | 4      | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
|            | b.     | Describe briefly the construction and working of coagulation sedimentation tank with neat sketch.                                                                                                                                                                                                                                                                                                   | 10     | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
|            | c.     | Briefly explain the terms : i) Sedimentation ii) Coagulation<br>iii) Flocculation.                                                                                                                                                                                                                                                                                                                  | 6      | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
| OR         |        |                                                                                                                                                                                                                                                                                                                                                                                                     |        |        |        |      |      |      |            |        |        |        |        |        |    |    |     |
| Q.4        | a.     | The maximum daily demand at a water purification plant has been estimated as 12 million litres per day. Design the dimensions of a suitable sedimentation tank (fitted with mechanical sludge removal arrangements) for the raw supplies , assuming a detention period of hours and velocity of flow as 20cm per minute.                                                                            | 8      | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
|            | b.     | List the coagulants used in water treatment.                                                                                                                                                                                                                                                                                                                                                        | 4      | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
|            | c.     | Briefly explain the mechanism of Filtration.                                                                                                                                                                                                                                                                                                                                                        | 8      | L2     | CO2    |      |      |      |            |        |        |        |        |        |    |    |     |
| Module – 3 |        |                                                                                                                                                                                                                                                                                                                                                                                                     |        |        |        |      |      |      |            |        |        |        |        |        |    |    |     |
| Q.5        | a.     | What is Disinfection of water? What are the characteristics of good disinfectant?                                                                                                                                                                                                                                                                                                                   | 6      | L2     | CO3    |      |      |      |            |        |        |        |        |        |    |    |     |

|                   |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
|                   | <b>b.</b> | Explain the different types of sewerage system with their merits and demerits of suitability.                                                                                                                                                                                                   | <b>12</b> | <b>L2</b> | <b>CO3</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
| <b>Q.6</b>        | <b>a.</b> | Calculate the velocity of flow and discharge of sewer of a circular section having a diameter of 1m laid at a gradient of 1 in 500. Use Manning's formula taking $N = 0.012$ . Assume that the sewer is running half full.                                                                      | <b>8</b>  | <b>L3</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain the process and objective of sampling with different methods.                                                                                                                                                                                                                           | <b>6</b>  | <b>L2</b> | <b>CO3</b> |
|                   | <b>c.</b> | Explain DWF and WWF.                                                                                                                                                                                                                                                                            | <b>6</b>  | <b>L2</b> | <b>CO3</b> |
| <b>Module – 4</b> |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
| <b>Q.7</b>        | <b>a.</b> | Illustrate the layout of a conventional municipal treatment plant and infer upon importance of each unit in sanitation.                                                                                                                                                                         | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | Elucidate the working principle of sludge digester, with a neat labeled sketch.                                                                                                                                                                                                                 | <b>6</b>  | <b>L2</b> | <b>CO4</b> |
|                   | <b>c.</b> | Explain different types of screens.                                                                                                                                                                                                                                                             | <b>4</b>  | <b>L2</b> | <b>CO4</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
| <b>Q.8</b>        | <b>a.</b> | Discuss briefly with a neat sketch Grit Chamber and Oil and Grease removal tank.                                                                                                                                                                                                                | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | Explain the working of Conventional Activated Sludge Process (ASP) with flow diagram.                                                                                                                                                                                                           | <b>10</b> | <b>L2</b> | <b>CO4</b> |
| <b>Module – 5</b> |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
| <b>Q.9</b>        | <b>a.</b> | Determine the size of a high rate trickling filter for the following data :<br>i) Sewage flow = 4.5 m <sup>3</sup> /d      ii) Recirculation Ratio = 1.5<br>iii) BOD of raw sewage = 250 mg/ℓ.<br>iv) BOD Removal in primary settling tank = 30%.<br>v) Final effluent BOD desired = 30 mg/ℓtr. | <b>10</b> | <b>L3</b> | <b>CO5</b> |
|                   | <b>b.</b> | Explain the concept of BoD and CoD. Enumerate their limitation.                                                                                                                                                                                                                                 | <b>6</b>  | <b>L2</b> | <b>CO5</b> |
|                   | <b>c.</b> | Briefly explain Self – Cleansing Velocity.                                                                                                                                                                                                                                                      | <b>4</b>  | <b>L2</b> | <b>CO5</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                 |           |           |            |
| <b>Q.10</b>       | <b>a.</b> | Draw a neat sketch of Skimming tank. Enumerate importance of Skimming tank.                                                                                                                                                                                                                     | <b>10</b> | <b>L2</b> | <b>CO5</b> |
|                   | <b>b.</b> | Draw and explain Oxidation Pond and Oxidation ditch.                                                                                                                                                                                                                                            | <b>10</b> | <b>L2</b> | <b>CO5</b> |

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Question Paper Version : A

## Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

### INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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- 
1. Biodiversity is a measure of variation at the \_\_\_\_\_ level  
 a) Genetic                      b) Species                      c) Ecosystem                      d) All of these
  2. World Environment Day is celebrated on \_\_\_\_\_  
 a) 5<sup>th</sup> May                      b) 5<sup>th</sup> June                      c) 18<sup>th</sup> June                      d) 16<sup>th</sup> August
  3. Mining means  
 a) To conserve minerals  
 b) To check pollution  
 c) To extract minerals and ores  
 d) None of the above
  4. Direct conversion of solar energy is attained by  
 a) Solar Photo volcanic system  
 b) Solar diesel hybrid system  
 c) Solar thermal system  
 d) Solar air heater
  5. What % of its geographical area of a country should be under forest cover  
 a) 23%                      b) 43%                      c) 13%                      d) 33%
  6. Hazardous Waste Management Act was enacted in India in the year  
 a) 1988                      b) 1989                      c) 1990                      d) 1991
  7. Which of these following elements is the case of e-waste?  
 a) Cadmium                      b) Beryllium                      c) Lead                      d) All of these

8. Remote sensing techniques make use of the properties of \_\_\_\_\_ emitted, reflected or diffracted by the sensed objects
  - a) Electric waves
  - b) Sound waves
  - c) Electromagnetic waves
  - d) Wind waves
9. The altitudinal distance of a geostationary satellite from the earth is about
  - a) 26,000km
  - b) 30,000km
  - c) 36,000km
  - d) 44,000km
10. Montreal protocol is related to the
  - a) Food security
  - b) Global warming
  - c) Sustainable development
  - d) Ozone layer depletion
11. The main cause of damage to Taj Mahal is \_\_\_\_\_
  - a) Water pollution
  - b) Soil pollution
  - c) Acid rain
  - d) Fog
12. Reducing the amount of future climate change is called.
  - a) Mitigation
  - b) Geo-engineering
  - c) Adaptation
  - d) None of these
13. Ozone layer is at a height of \_\_\_\_\_ above the Earth's surface
  - a) 19 to 48m
  - b) 19 to 480m
  - c) 19 to 48km
  - d) 190 to 480km
14. Which ministry is mainly responsible for research and development in renewable energy sources such as wind, power small hydro, biogas and solar power
  - a) Human Resource Development
  - b) Agriculture and Famous welfare
  - c) Ministry of new and Renewable energy
  - d) Health and Family welfare
15. The OTEC is an energy technology that converts
  - a) Energy in large fides of ocean to generate electricity
  - b) Energy in ocean waves to generate electricity
  - c) Energy in ocean due to thermal gradient to generate electricity
  - d) Energy in the fast moving ocean currents to generate electricity
16. In a Lake, phytoplankton grow in abundance in
  - a) Littoral zone
  - b) Limnetic zone
  - c) Profundal zone
  - d) Benthic region
17. The prescribed limits of noise in residential area during day is
  - a) 55dB
  - b) 45dB
  - c) 60dB
  - d) 50dB
18. The maximum allowable concentration of fluorides in drinking water
  - a) 3mg/L
  - b) 2mg/L
  - c) 2.5mg/L
  - d) 1.5mg/L
19. The color code of plastic bag for disposing microbial laboratory culture waste
  - a) Red
  - b) Black
  - c) Blue
  - d) White
20. The hazardous pollutant released from batteries is
  - a) Arsenic
  - b) Cobalt
  - c) Barium
  - d) Cadmium
21. GIS uses the information from which of the following sources
  - a) Non-Spatial Information System
  - b) Spatial Information System
  - c) Global Information System
  - d) Position Information System

22. EIA can be expanded as  
 a) Environment and Industrial Act  
 b) Environmental and Impact Activities  
 c) Environmental Impact Assessment  
 d) Environmental Impact Activity
23. ISO 14000 standards deals with  
 a) Pollution management  
 b) Risk management  
 c) Environmental management  
 d) None of these
24. Which of the following represents India in ISO  
 a) PFRDA  
 b) FSSAI  
 c) BIS  
 d) BCCI
25. Which of the following is having high population density  
 a) India  
 b) China  
 c) USA  
 d) Western Europe
26. Environment education is targeted to  
 a) General public  
 b) Professional social groups  
 c) Technical and Scientists  
 d) All of the above
27. Discharge of municipal waste cause  
 a) Depletion of dissolved oxygen  
 b) Destroy aquatic life  
 c) Impair biological activity  
 d) All of the above
28. \_\_\_\_\_ is are referred to a Earth's lungs  
 a) Forests  
 b) Carbon cycle  
 c) Water sources  
 d) Miner
29. Solid waste is best managed through  
 a) Incineration  
 b) Open dumping  
 c) Sanitary landfill  
 d) Composting
30. Love canal tragedy is attributed to  
 a) Soil pollution  
 b) Hazardous waste  
 c) Air pollution  
 d) None of these
31. According to Biomedical Waste (Management and Handling) Rules 1998, waste should not be stored beyond \_\_\_\_\_  
 a) 12 hours  
 b) 48 hours  
 c) 72 hours  
 d) 96 hours
32. Pyrolysis is an \_\_\_\_\_ process  
 a) Exothermic  
 b) Endothermic  
 c) Both a and b  
 d) Neither a and b
33. Chlorofluorocarbons are  
 a) Nontoxic  
 b) Flammable  
 c) Corrosive  
 d) Odorous
34. Which of the following is an air pollutant  
 a) Carbon dioxide  
 b) Oxygen  
 c) Nitrogen  
 d) Particulate matter
35. Urbanization is  
 a) Local environmental issue  
 b) Nation environmental issue  
 c) Both a and b  
 d) Not at all an issue

36. Earth day is held every year on :  
 a) June 5<sup>th</sup>                      b) November 23<sup>rd</sup>                      c) April 22<sup>nd</sup>                      d) January 26<sup>th</sup>
37. The term hotspot was introduced by –  
 a) Norman Myere                      b) Jacob Von Verkul  
 c) A.G. Transley                      d) Ernst Haeckel
38. In an Ecosyste, the energy flow is always  
 a) Always unidirectional                      b) Always bidirectional  
 c) In any direction                      d) Always down directional
39. Which of the following is considered as an alternate fuel  
 a) CNG                      b) Kerosene                      c) Coal                      d) Petrol
40. Nuclear power plant in Karnataka is located at  
 a) Bhadravati                      b) Sandur                      c) Raichur                      d) Kaiga
41. Disaster management includes \_\_\_\_\_  
 a) Mitigation                      b) Reconstruction                      c) Rehabilitation                      d) All of these
42. What is the health effects of fluoride in drinking waster  
 a) Arthritis                      b) Diarrhea                      c) Anemia                      d) All of these
43. What is the permissible range of pH for drinking water as per Indian standards  
 a) 6 – 9                      b) 6 – 8.5                      c) 6.5 – 8.5                      d) 6 5 – 7.5
44. The infiltration of water into the subsurface is the  
 a) Influent                      b) Effluent                      c) Discharge                      d) Recharge
45. Environmental (Protection Act) was enacted in the year  
 a) 1986                      b) 1992                      c) 1984                      d) 1974
46. What is the full form of NGO  
 a) Non-Governmental Organization  
 b) Non-Governance Organizations  
 c) No- Governance Organizations  
 d) Null – Governmental organizations
47. The primary cause of acid rain around the world is  
 a) CFC                      b) SO<sub>2</sub>                      c) CO                      d) O<sub>3</sub>
48. Bhopal Gas Tragedy caused due to leakage of  
 a) Methyl ISO Cyanate (MIC)                      b) Sulphur dioxide  
 c) Mustered gas                      d) Methane
49. Deforestation can  
 a) Increase the rainfall                      b) Increase soil fertility  
 c) Introduce silt in the river                      d) None of these
50. The word Environment is derived from  
 a) Greek                      b) French                      c) Spanish                      d) English

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Question Paper Version : B

**Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024**  
**Environmental Studies**

Time: 1 hr.]

[Max. Marks: 50

**INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
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- 
- 1 Disaster management includes \_\_\_\_\_  
a) Mitigation                      b) Reconstruction                      c) Rehabilitation                      d) All of these
  - 2 What is the health effects of fluoride in drinking waster  
a) Arthritis                      b) Diarrhea                      c) Anemia                      d) All of these
  - 3 What is the permissible range of pH for drinking water as per Indian standards  
a) 6 – 9                      b) 6 – 8.5                      c) 6.5 – 8.5                      d) 6.5 – 7.5
  - 4 The infiltration of water into the subsurface is the  
a) Influent                      b) Effluent                      c) Discharge                      d) Recharge
  - 5 Environmental (Protection Act) was enacted in the year  
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 a) Always unidirectional  
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 b) 19 to 480m  
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Question Paper Version : C

**Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024**  
**Environmental Studies**

Time: 1 hr.]

[Max. Marks: 50

**INSTRUCTIONS TO THE CANDIDATES**

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  - 5 Urbanization is  
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 a) Soil pollution                      b) Hazardous waste                      c) Air pollution                      d) None of these
- 41 The main cause of damage to Taj Mahal is \_\_\_\_\_  
 a) Water pollution                      b) Soil pollution                      c) Acid rain                      d) Fog
- 42 Reducing the amount of future climate change is called.  
 a) Mitigation                      b) Geo-engineering                      c) Adaptation                      d) None of these
- 43 Ozone layer is at a height of \_\_\_\_\_ above the Earth's surface  
 a) 19 to 48m                      b) 19 to 480m                      c) 19 to 48km                      d) 190 to 480km
- 44 Which ministry is mainly responsible for research and development in renewable energy sources such as wind, power small hydro, biogas and solar power  
 a) Human Resource Development  
 b) Agriculture and Famous welfare  
 c) Ministry of new and Renewable energy  
 d) Health and Family welfare
- 45 The OTEC is an energy technology that converts  
 a) Energy in large fides of ocean to generate electricity  
 b) Energy in ocean waves to generate electricity  
 c) Energy in ocean due to thermal gradient to generate electricity  
 d) Energy in the fast moving ocean currents to generate electricity
- 46 In a Lake, phytoplankton grow in abundance in  
 a) Littoral zone                      b) Limnetic zone                      c) Profundal zone                      d) Benthic region
- 47 The prescribed limits of noise in residential area during day is  
 a) 55dB                      b) 45dB                      c) 60dB                      d) 50dB
- 48 The maximum allowable concentration of fluorides in drinking water  
 a) 3mg/L                      b) 2mg/L                      c) 2.5mg/L                      d) 1.5mg/L
- 49 The color code of plastic bag for disposing microbial laboratory culture waste  
 a) Red                      b) Black                      c) Blue                      d) White
- 50 The hazardous pollutant released from batteries is  
 a) Arsenic                      b) Cobalt                      c) Barium                      d) Cadmium

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Question Paper Version : D

**Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024**  
**Environmental Studies**

Time: 1 hr.]

[Max. Marks: 50

**INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
- 1 GIS uses the information from which of the following sources
    - a) Non-Spatial Information System
    - b) Spatial Information System
    - c) Global Information System
    - d) Position Information System
  - 2 EIA can be expanded as
    - a) Environment and Industrial Act
    - b) Environmental and Impact Activities
    - c) Environmental Impact Assessment
    - d) Environmental Impact Activity
  - 3 ISO 14000 standards deals with
    - a) Pollution management
    - b) Risk management
    - c) Environmental management
    - d) None of these
  - 4 Which of the following represents India in ISO
    - a) PFRDA
    - b) FSSAI
    - c) BIS
    - d) BCCI
  - 5 Which of the following is having high population density
    - a) India
    - b) China
    - c) USA
    - d) Western Europe
  - 6 Environment education is targeted to
    - a) General public
    - b) Professional social groups
    - c) Technical and Scientists
    - d) All of the above

- 7 Discharge of municipal waste cause  
 a) Depletion of dissolved oxygen  
 b) Destroy aquatic life  
 c) Impair biological activity  
 d) All of the above
- 8 \_\_\_\_\_ is are referred to a Earth's lungs  
 a) Forests                      b) Carbon cycle                      c) Water sources                      d) Miner
- 9 Solid waste is best managed through  
 a) Incineration                      b) Open dumping                      c) Sanitary landfill                      d) Composting
- 10 Love canal tragedy is attributed to  
 a) Soil pollution                      b) Hazardous waste                      c) Air pollution                      d) None of these
- 11 Disaster management includes \_\_\_\_\_  
 a) Mitigation                      b) Reconstruction                      c) Rehabilitation                      d) All of these
- 12 What is the health effects of fluoride in drinking waster  
 a) Arthritis                      b) Diarrhea                      c) Anemia                      d) All of these
- 13 What is the permissible range of pH for drinking water as per Indian standards  
 a) 6 – 9                      b) 6 – 8.5                      c) 6.5 – 8.5                      d) 6.5 – 7.5
- 14 The infiltration of water into the subsurface is the  
 a) Influent                      b) Effluent                      c) Discharge                      d) Recharge
- 15 Environmental (Protection Act) was enacted in the year  
 a) 1986                      b) 1992                      c) 1984                      d) 1974
- 16 What is the full form of NGO  
 a) Non-Governmental Organization  
 b) Non-Governance Organizations  
 c) No- Governance Organizations  
 d) Null – Governmental organizations
- 17 The primary cause of acid rain around the world is  
 a) CFC                      b) SO<sub>2</sub>                      c) CO                      d) O<sub>3</sub>
- 18 Bhopal Gas Tragedy caused due to leakage of  
 a) Methyl ISO Cyanate (MIC)                      b) Sulphur dioxide  
 c) Mustered gas                      d) Methane
- 19 Deforestation can  
 a) Increase the rainfall  
 b) Increase soil fertility  
 c) Introduce silt in the river  
 d) None of the above
- 20 The word Environment is derived from  
 a) Greek                      b) French                      c) Spanish                      d) English
- 21 According to Biomedical Waste (Management and Handling) Rules 1998, waste should not be stored beyond \_\_\_\_\_  
 a) 12 hours                      b) 48 hours                      c) 72 hours                      d) 96 hours

- 22 Pyrolysis is an \_\_\_\_\_ process  
a) Exothermic      b) Endothermic      c) Both a and b      d) Neither a and b
- 23 Chlorofluorocarbons are  
a) Nontoxic      b) Flammable      c) Corrosive      d) Odorous
- 24 Which of the following is an air pollutant  
a) Carbon dioxide      b) Oxygen      c) Nitrogen      d) Particulate matter
- 25 Urbanization is  
a) Local environmental issue  
b) Nation environmental issue  
c) Both a and b  
d) Not at all an issue
- 26 Earth day is held every year on :  
a) June 5<sup>th</sup>      b) November 23<sup>rd</sup>      c) April 22<sup>nd</sup>      d) January 26<sup>th</sup>
- 27 The term hotspot was introduced by –  
a) Norman Myere      b) Jacob Von Verkul  
c) A.G. Transley      d) Ernst Haeckel
- 28 In an Ecosystem, the energy flow is always  
a) Always unidirectional  
b) Always bidirectional  
c) In any direction  
d) Always down directional
- 29 Which of the following is considered as an alternate fuel  
a) CNG      b) Kerosene      c) Coal      d) Petrol
- 30 Nuclear power plant in Karnataka is located at  
a) Bhadravati      b) Sandur      c) Raichur      d) Kaiga
- 31 The main cause of damage to Taj Mahal is \_\_\_\_\_  
a) Water pollution      b) Soil pollution      c) Acid rain      d) Fog
- 32 Reducing the amount of future climate change is called.  
a) Mitigation      b) Geo-engineering      c) Adaptation      d) None of these
- 33 Ozone layer is at a height of \_\_\_\_\_ above the Earth's surface  
a) 19 to 48m      b) 19 to 480m      c) 19 to 48km      d) 190 to 480km
- 34 Which ministry is mainly responsible for research and development in renewable energy sources such as wind, power small hydro, biogas and solar power  
a) Human Resource Development  
b) Agriculture and Famous welfare  
c) Ministry of new and Renewable energy  
d) Health and Family welfare
- 35 The OTEC is an energy technology that converts  
a) Energy in large fides of ocean to generate electricity  
b) Energy in ocean waves to generate electricity  
c) Energy in ocean due to thermal gradient to generate electricity  
d) Energy in the fast moving ocean currents to generate electricity

- 36 In a Lake, phytoplankton grow in abundance in  
a) Littoral zone      b) Limnetic zone      c) Profundal zone      d) Benthic region
- 37 The prescribed limits of noise in residential area during day is  
a) 55dB      b) 45dB      c) 60dB      d) 50dB
- 38 The maximum allowable concentration of fluorides in drinking water  
a) 3mg/L      b) 2mg/L      c) 2.5mg/L      d) 1.5mg/L
- 39 The color code of plastic bag for disposing microbial laboratory culture waste  
a) Red      b) Black      c) Blue      d) White
- 40 The hazardous pollutant released from batteries is  
a) Arsenic      b) Cobalt      c) Barium      d) Cadmium
- 41 Biodiversity is a measure of variation at the \_\_\_\_\_ level  
a) Genetic      b) Species      c) Ecosystem      d) All of these
- 42 World Environment Day is celebrated on \_\_\_\_\_  
a) 5<sup>th</sup> May      b) 5<sup>th</sup> June      c) 18<sup>th</sup> June      d) 16<sup>th</sup> August
- 43 Mining means  
a) To conserve minerals      b) To check pollution  
c) To extract minerals and ores      d) None of these
- 44 Direct conversion of solar energy is attained by  
a) Solar Photo volcanic system  
b) Solar diesel hybrid system  
c) Solar thermal system  
d) Solar air heater
- 45 What % of its geographical area of a country should be under forest cover  
a) 23%      b) 43%      c) 13%      d) 33%
- 46 Hazardous Waste Management Act was enacted in India in the year  
a) 1988      b) 1989      c) 1990      d) 1991
- 47 Which of these following elements is the case of e-waste?  
a) Cadmium      b) Beryllium      c) Lead      d) All of these
- 48 Remote sensing techniques make use of the properties of \_\_\_\_\_ emitted, reflected or diffracted by the sensed objects  
a) Electric waves      b) Sound waves  
c) Electromagnetic waves      d) Wind waves
- 49 The altitudinal distance of a geostationary satellite from the earth is about  
a) 26,000km      b) 30,000km      c) 36,000km      d) 44,000km
- 50 Montreal protocol is related to the  
a) Food security      b) Global warming  
c) Sustainable development      d) Ozone layer depletion

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## Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Hydrology and Water Resources Engineering

Time: 3 hrs.

Max. Marks: 100

**Note: Answer any FIVE full questions, choosing ONE full question from each module.**

### Module-1

- 1 a. Explain Hydrology – an Interdisciplinary Science. What are the practical application of hydrology? (10 Marks)
- b. The analysis of a storm yielded the following :

| Isohyet Interval (mm)  | 70-80 | 80-90 | 90-100 | 100-110 | 110-120 | 120-130 |
|------------------------|-------|-------|--------|---------|---------|---------|
| Area (km) <sup>2</sup> | 10    | 85    | 113    | 98      | 136     | 67      |

Calculate the average depth of rainfall. (05 Marks)

- c. Explain Double Mass curve technique to test consistency of data. (05 Marks)

**OR**

- 2 a. With a neat sketch, explain Horton's Qualitative representation of Hydrologic cycle. List the different forms of precipitation. (08 Marks)
- b. The average rainfall in cm at four existing rain gauge stations in a basin are 105, 79, 70 and 66. If the average depth of rainfall over basin is to be estimated with 10% error, determine the additional number of rain gauge required. (07 Marks)
- c. Explain with a neat sketch, Syphon type recording Rain gauge. (05 Marks)

### Module-2

- 3 a. What are the factors affecting Evaporation rate? Explain briefly. (08 Marks)
- b. What are the components of unit hydrograph? Write a note on its applications. (06 Marks)
- c. Define Unit hydrograph and what are assumptions made the unit hydrograph. (06 Marks)

**OR**

- 4 a. What are the factors affecting the Runoff and explain details. (08 Marks)
- b. What are the limitations of Unit hydrograph theory? (06 Marks)
- c. Following are the ordinate of a 3 hours unit hydrograph derive the plot the 3 hours flood hydrograph due to an excess rainfall of 4.5 cms. (06 Marks)

| Time (Hours)                    | 0 | 3   | 6   | 9   | 12 | 15  | 18  | 21  | 24  |
|---------------------------------|---|-----|-----|-----|----|-----|-----|-----|-----|
| 3 – Hrs UHG m <sup>3</sup> /sec | 0 | 1.5 | 4.5 | 8.6 | 12 | 9.4 | 4.6 | 2.3 | 0.8 |

### Module-3

- 5 a. Define Irrigation. What is the necessity of Irrigation? (06 Marks)
- b. Explain the surface irrigation methods of water application to the crops. (08 Marks)
- c. Compare Flow irrigation and Lift irrigation. (06 Marks)

**OR**

- 6 a. Define Duty, Delta and Base period and establish a relationship between them. (08 Marks)
- b. What are factors affecting duty? What are the methods to improve the duty? (06 Marks)
- c. Explain the following terms : i) Application efficiency ii) Conveyance efficiency  
iii) Frequency of irrigation. (06 Marks)

**Module-4**

- 7 a. With neat sketch, explain the types of canal alignment. (10 Marks)  
b. List the various factors to be considered in selecting a site for a reservoir. (10 Marks)

**OR**

- 8 a. Explain the Mass curve method to determine the storage capacity of Reservoir. (12 Marks)  
b. What are the advantages of Lacey's theory compared to Kennedy's theory? (08 Marks)

**Module-5**

- 9 a. Which Indian rivers cause the most floods in India? (10 Marks)  
b. Explain the importance of water harvesting and conservation along with basic principles involved in process. (10 Marks)

**OR**

- 10 a. What are the steps taken to control drought? (10 Marks)  
b. Define the term "Rain Water Harvesting". Elaborate Rural technological system being adopted for water conservation. (10 Marks)

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## Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Transportation Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: Answer any FIVE full questions, choosing ONE full question from each module.*

### Module-1

- 1 a. Explain the importance of transportation. Write the advantages and disadvantages of different modes of transportation. (10 Marks)
- b. Explain interpretation of planning survey. Briefly explain Vision 2021. (10 Marks)

OR

- 2 a. Explain Engineering Surveys. (10 Marks)
- b. Explain briefly various stages of work in new highway project. (10 Marks)

### Module-2

- 3 a. A state highway passing through a rolling terrain has a horizontal curve of radius equal to ruling minimum radius. Design all geometric features of horizontal curve, assuming suitable data. (12 Marks)
- b. With a neat sketch briefly explain about camber and width of carriage way. (08 Marks)

OR

- 4 a. Find safe overtaking sight distance for design speed 96 kmph, acceleration of overtaking vehicle 1.92 kmph/sec, draw neat sketch of overtaking zone (minimum). (10 Marks)
- b. Write the difference between flexible pavement and rigid pavement. (10 Marks)

### Module-3

- 5 a. Explain the properties of aggregates and tests to be conducted on aggregates. (10 Marks)
- b. CBR tests were conducted on two specimens of a soil. Determine CBR value of the soil, if 100 divisions of load dial represents 190kg load in the calibration chart of the proving ring.

| Penetration of plunger, mm | Load dial reading, divisions |            |
|----------------------------|------------------------------|------------|
|                            | Specimen 1                   | Specimen 2 |
| 0.0                        | 0                            | 0          |
| 0.5                        | 8                            | 0.5        |
| 1.0                        | 15                           | 1.5        |
| 1.5                        | 23                           | 2.5        |
| 2.0                        | 29                           | 6.0        |
| 2.5                        | 34                           | 13         |
| 3.0                        | 37                           | 20         |
| 4.0                        | 43                           | 30         |
| 5.0                        | 48                           | 38         |
| 7.5                        | 57                           | 50         |
| 10.0                       | 63                           | 58         |
| 12.5                       | 67                           | 63         |
| -                          | -                            | -          |

(10 Marks)

**OR**

- 6 a. Explain the design elements of highway embankment. (10 Marks)  
b. Explain in brief the specification of material and construction steps of WMM layer. (10 Marks)

**Module-4**

- 7 a. With a neat sketch, explain the procedure for design of filter material in highway drainage. (10 Marks)  
b. Explain in brief three methods of economic evaluation of highway projects. (10 Marks)

**OR**

- 8 a. Explain in briefly highway user benefits and Vehicle Operation Cost (VOC). (10 Marks)  
b. Write note on BOT and BOOT. (10 Marks)

**Module-5**

- 9 a. Explain about traffic regulation and controls. (10 Marks)  
b. Define the term "Runway Orientation". Explain about any one type of wind rose diagram. (10 Marks)

**OR**

- 10 a. Explain briefly functions of sleepers and ballast. (10 Marks)  
b. Explain factors affecting selection of sites for an airport. (10 Marks)

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## Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Design of RC Structural Elements

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. Use of IS-456-2000, Sp-16 is permitted.  
3. Assume suitable additional data, if necessary.

### Module-1

- 1 a. Distinguish between working stress and limit state method of RCC design. (06 Marks)
- b. Explain balanced, under-reinforced and over-reinforced section with sketches. (06 Marks)
- c. From fundamental, derive expression for the area of stress block  $0.36f_{ck}x_n$ . (08 Marks)

**OR**

- 2 a. Explain the factors that affect short and long term deflection. (06 Marks)
- b. A simply supported beam of rectangular section 200mm wide by 450mm overall depth is used over an effective span of 5m. The beam is reinforced with 3 bars of 16mm diameter of an effective depth of 420mm. the beam supports a live load of 10kN/m. Calculate short term deflection using M20 concrete and Fe 415 steel. (14 Marks)

### Module-2

- 3 a. A singly reinforced beam of 250mm wide by 500mm effective depth is reinforced with 4 bars of 22mm diameter. Effective span of beam is 6m. Assuming M20 grade concrete and Fe415 steel, determine the concentrated load at midspan that can be carried by the beam in addition to its self weight. (12 Marks)
- b. A T-beam of depth 450mm has a flange width of 1000mm and depth of 120mm. It is reinforced with 6 bars of 20mm diameter on tension side with a cover of 30mm. If M20 concrete and Fe415 steel are used, calculate the moment of resistance of the beam. Take web width = 300mm. (08 Marks)

**OR**

- 4 a. A doubly reinforced beam section is of 250mm wide and 500mm deep to the center of the tensile reinforcement. It is reinforced with 2 bars of 16mm diameter as compression steel at an effective cover of 40mm and 4 bars of 25mm diameter as tensile steel. Using M20 grade concrete and Fe415 steel, calculate the factored moment of resistance. (12 Marks)
- b. Determine the area of tensile reinforcement required for T-beam having following details  
Effective flange width = 2400mm  
Depth of flange = 150mm  
Width of web = 300mm  
Effective depth = 750mm  
Working moment 800kN-m  
Type of concrete = M20 ; Type of steel = Fe415 MYSD bars (08 Marks)

### Module-3

- 5 Design a cantilever beam of clear span 3.25m, service load is 15kN-m. Use M20 grade concrete and Fe415 steel. Sketch the reinforcement details. (20 Marks)

OR

- 6 a. List the circumstances under which doubly reinforced beams are recommended. (04 Marks)  
 b. A RC beam is supported on two walls of 500mm thick, spaced at a clear distance of 6m. The beam carries a super imposed load of 30kN/m. Use M20 grade concrete and Fe415 steel. Design the beam for flexure and shear. Sketch the reinforcement detail. (16 Marks)

**Module-4**

- 7 The slab of a residential building of size  $4.3\text{m} \times 6\text{m}$  is simply supported on all four sides on 230mm walls. Assuming an imposed load of  $2\text{kN/m}^2$ , and load due to finishes of  $1.0\text{kN/m}^2$ , design the floor slab. Use M25 concrete and Fe 415 steel. Assume mild exposure. Sketch the reinforcement details. (20 Marks)

OR

- 8 Design one of the flights of a dog-legged staircase for an office building, given the following data :  
 Height between floor = 3.2m, Riser = 160mm, Tread = 270mm, Width of flight = loading width = 1.25m, Live load =  $5\text{kN/m}^2$ , finishes load =  $0.6\text{kN/m}^2$ . Assume the stairs to be supported on 230mm thick masonry walls at the outer edges of the landing, parallel to the risers. Use M20 concrete and Fe 415 steel. Assume mild exposure conditions. Sketch the reinforcement details. (20 Marks)

**Module-5**

- 9 a. What is the note of transverse reinforcement in columns? What are the codal provisions to design the transverse reinforcement? (05 Marks)  
 b. Design a short reinforced concrete column of rectangular section to carry an ultimate load of 600kN and ultimate moment of 100kN-m, acting about an axis bisecting the depth of the column. Assume the effective length of column is 4.5m. Width of the supported beam is 300mm. Use M20 concrete and Fe415 steel, provide equal steel on both tension and compression sides. Sketch reinforcement detail. (15 Marks)

OR

- 10 Design an isolated footing for a square column of size  $400\text{mm} \times 400\text{mm}$ , supporting a service load of 2200kN. Assume SBC of soil as  $250\text{ kN/m}^2$  at a depth of 1.5m below the ground. Use M20 concrete and Fe 415 steel for the footing. Sketch the reinforcement details. (20 Marks)

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# CBCS SCHEME

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## Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

**Note: Answer any FIVE full questions, choosing ONE full question from each module.**

### Module-1

- 1 a. With the help of 3-phase diagram, explain:  
(i) Void ratio (ii) Porosity (iii) Air content (iv) Degree of saturation (10 Marks)
- b. With usual notations, derive the relation  $Y_d = \frac{(1 - n_a)GY_w}{WG + 1}$ . (10 Marks)

**OR**

- 2 a. Explain the following:  
(i) Liquid limit (ii) Shrinkage limit (iii) Plastic limit  
(iv) Activity of clay (v) Relative density (10 Marks)
- b. Classify following soils as per IS classification:

| Soil | % passing 4.75 mm | % passing 75 micron | $C_u$ | $C_c$ | W                         |
|------|-------------------|---------------------|-------|-------|---------------------------|
| A    | 98                | 02                  | 6.7   | 1.2   | $I_p = 2.2$               |
| B    | 100               | 100                 | -     | -     | $W_L = 400$<br>$W_p = 50$ |

(10 Marks)

### Module-2

- 3 a. Define permeability and explain factors affecting permeability. (10 Marks)
- b. The discharge of water collected from a constant head permeameter in a period of 15 minutes is 400 ml. The internal diameter of permeameter is 6cm and measured difference in heads between the two gauging points 15 cm apart is 40. Calculate the coefficient of permeability. If the dry weight of 15 cm long sample is 7N and specific gravity of the solids is 2.65. Calculate seepage velocity. (10 Marks)

**OR**

- 4 a. Explain effective stress and total stress concept. (08 Marks)
- b. Write a note on quick sand phenomena. (06 Marks)
- c. Explain impact of effective stress in construction of structures. (06 Marks)

### Module-3

- 5 a. Discuss the factors affecting compaction of soils. (05 Marks)
- b. Write difference between standard and modified proctor's compaction. (03 Marks)
- c. The following are the results of compaction test:

|                       |      |      |      |      |      |      |
|-----------------------|------|------|------|------|------|------|
| Water content (%)     | 8    | 11.5 | 14.5 | 17.5 | 19.5 | 21.5 |
| Mass of wet soil (gm) | 1700 | 1900 | 2000 | 1980 | 1950 | 1920 |

If volume of mould used was 950 cc and specific gravity was 2.65:

- (i) Draw dry density versus water content curve, find OMC and MDD.  
(ii) Plot 100% saturation line (12 Marks)

OR

- 6 a. Write assumptions and Terzaghi's one dimensional consolidation theory. (06 Marks)  
 b. With neat sketch, explain how pre-consolidation pressure is determined by Casagrande's method. (06 Marks)  
 c. A soil sample of 20 mm thick take 20 minutes to reach 20% consolidation. Find the time taken for a clay layer 6m thick to reach 40% consolidation. Assume double drainage in both cases. (08 Marks)

**Module-4**

- 7 a. Explain Mohr's Coulomb's theory as applied to soil strength. (06 Marks)  
 b. Write note on total and effective shear strength. (06 Marks)  
 c. A specimen of clean dry, cohesionless sand is tested in shear box and soil is failed at shear stress of  $40 \text{ kN/m}^2$  when the normal load on the specimen was  $50 \text{ kN/m}^2$ . Determine:  
 (i) Angle of shearing resistance  
 (ii) The principal stresses during the failure (08 Marks)

OR

- 8 a. List the factors affecting shear strength of soils. (04 Marks)  
 b. Explain Thixotropy and sensitivity. (06 Marks)  
 c. Two identical soil specimens were tested in triaxial apparatus. The first specimen failed at a deviator stress of  $770 \text{ kN/m}^2$ . When the cell pressure was  $200 \text{ kN/m}^2$ . Second specimen failed at deviator stress of  $1370 \text{ kN/m}^2$  under pressure of  $400 \text{ kN/m}^2$ . Determine the value of cohesion and angle of internal friction analytically. If soil is tested in a direct shear apparatus with normal stress of  $600 \text{ kN/m}^2$ . Estimate shear stress at failure. (10 Marks)

**Module-5**

- 9 a. Explain modes of shear failure. (06 Marks)  
 b. Discuss effect of water table on bearing capacity with neat sketch. (06 Marks)  
 c. Compute the SBC of continuous footing at 1.8 m wide and locate at depth of 1.2 m below ground level in soil. Unit weight is  $20 \text{ kN/m}^3$  and cohesion  $20 \text{ kN/m}^2$  and internal friction  $20^\circ$ . Assume F.S. = 2.5. When is the permissible load per meter run of footing? Given  $N_c = 17.7$ ,  $N_g = 7.4$ ,  $N_y = 5$  (08 Marks)

OR

- 10 a. What are the different types of settlements of footing? Explain. (06 Marks)  
 b. Define differential and total settlement. (04 Marks)  
 c. A soft normally consolidated clay layer is 18 m thick. The natural water content is 45%. The saturated unit weight is  $18 \text{ kN/m}^3$ . The grain specific gravity is 2.7 and liquid limit is 63%. The vertical stress increment at the center of layer due to foundation load is  $9 \text{ kN/m}^2$ . The ground water table is at the surface of the clay layer. Determine the settlement of foundation. (10 Marks)

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## Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Research Methodology & Intellectual Property Rights

Time: 3 hrs.

Max. Marks: 100

*Note: Answer any FIVE full questions, choosing ONE full question from each module.*

### Module-1

- 1 a. What is Engineering Research? What are the primary objectives of conducting research in engineering? (10 Marks)
- b. What are the various types of engineering research? Explain. (10 Marks)

**OR**

- 2 a. Explain Fabrication, Falsification and Plagiarism related to Engineering research. (10 Marks)
- b. What ethical considerations and responsibilities should be taken into account when determining authorship in Engineering research? (10 Marks)

### Module-2

- 3 a. How do researchers distinguish between new and existing knowledge during a literature review? (10 Marks)
- b. How can researchers effectively use search engines to find relevant literature in their fields? (10 Marks)

**OR**

- 4 a. What challenges do researchers commonly face when reading mathematical content or algorithm? (10 Marks)
- b. What is impact of Title and Keywords on Citations? Explain Citation based knowledge flow. (10 Marks)

### Module-3

- 5 a. What is definition of Intellectual Property (IP)? In what way does Intellectual Property contribute to economic growth and cultural development in a society? (10 Marks)
- b. Discuss the history of Intellectual property in India. (10 Marks)

**OR**

- 6 a. Explain the step by step process of obtaining a patent. From the initial idea to the grant of the patent. (10 Marks)
- b. What are the commonly used terms in the field of patenting and how do they contribute to effective communication in this domain. (10 Marks)

### Module-4

- 7 a. Explain the criteria that an original work must meet to qualify for copyright protection. (10 Marks)
- b. Explain the process of copyright registration? What are the benefits for the copy right holders? (10 Marks)

**OR**

- 8 a. Explain the process of Trademark registration. (10 Marks)  
b. Explain the classification system for trademarks and its role in categorizing different types of marks. (10 Marks)

**Module-5**

- 9 a. Explain the process of Industrial design registration. (10 Marks)  
b. Explain the famous case law between Apple Inc Vs Samsung Electronics Co. related with Industrial Design rights. (10 Marks)

**OR**

- 10 a. Which specific acts, laws and rules govern geographical indications in India? Give some examples of well known geographical indications registered in India. (10 Marks)  
b. How would you describe the overall ecosystem and significance of geographical indications in India? (10 Marks)

\* \* \* \* \*



OR

- 2 What are the different types of Estimates? Explain any four different types of Estimates. (20 Marks)

**Module-2**

- 3 The details of man hole is as shown in Fig.Q.3. Estimate the quantities for the following item of work:
- Earthwork excavation in foundation
  - Cement concrete 1:3:6 floor and foundation
  - First class brick work with CM 1:4
  - 20mm thick cement plaster 1:3 in floor and channel.
- (20 Marks)

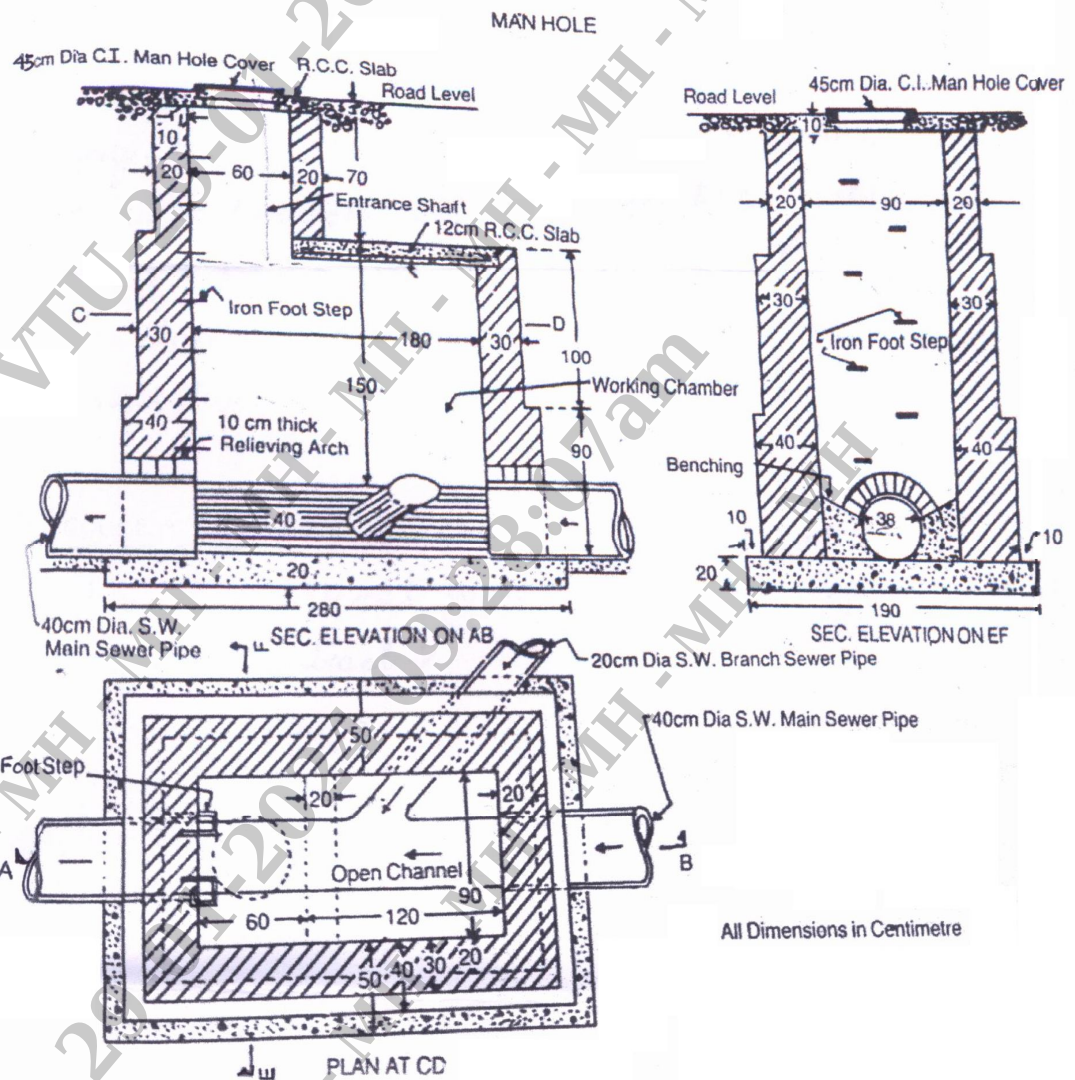


Fig.Q.3

OR

- 4 Estimate the quantity of earthwork in banking and cutting by mid sectional area method for a portion of road from the following data:

|                  |        |                                       |        |        |        |        |        |                               |        |        |        |        |        |
|------------------|--------|---------------------------------------|--------|--------|--------|--------|--------|-------------------------------|--------|--------|--------|--------|--------|
| Distance in 'm'  | 0      | 100                                   | 200    | 300    | 400    | 500    | 600    | 700                           | 800    | 900    | 1000   | 1100   | 1200   |
| R.L of ground    | 114.00 | 114.60                                | 115.00 | 115.20 | 116.10 | 116.50 | 118.00 | 118.25                        | 118.10 | 117.80 | 117.75 | 117.80 | 119.20 |
| R.L of formation | 115.0  | Upward gradient of 1 in 200 upto 600m |        |        |        |        |        | Downward gradient of 1 in 400 |        |        |        |        |        |

Formation width of road is 10m. Side slope 2:1 in banking and 1.5:1 in cutting. (20 Marks)

**Module-3**

- 5 Write detailed technical specification for the following:

- 12mm thick plastering for walls with CM1:6
- First class brick masonry for super structure CM1:4
- Earthwork excavation for foundation
- Mosaic/Terrazo flooring.

(20 Marks)

OR

- 6 Carryout the rate analysis for the following:

- Cement concrete CC 1:2:4 for RCC works.
- R.C.C.  $1:1\frac{1}{2}:3$  for roof slab.
- 1<sup>st</sup> class BBM in CM1:6 for superstructure.
- 20mm thick DPC with CM1:5.

(20 Marks)

**Module-4**

- 7 What are the different types of contract? Explain any four types of contracts?

(20 Marks)

OR

- 8 Explain the procedure of tendering and award of works in civil engineering projects.

(20 Marks)

**Module-5**

- 9 Write short notes about any four of the following:

- Performance security
- Breach of contract
- Mobilization and equipment advances
- Contract management
- Liquidated damages.

(20 Marks)

OR

- 10 a. Explain the methods of valuation. (10 Marks)
- b. A building of replacement value of about Rs.7,00,000/- stands on a main road on a leasehold plot. The ground rent per annum is Rs.2950/-. The building is of R.C.C. framed structure type. It is estimated that the building will have a future life of 70 years. The rent of the building is Rs.4000/- per month. The taxes payable are 18% of the gross rent and insurance premium is 0.5% of the gross rent. Assuming suitable figures for other items of the usual outgoings. Determine the capitalized value of the property on the basis of a 5% net yield. The S.F coefficient for the replacement of the capital is 70 years at 3% is 0.0043. (10 Marks)

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# CBCS SCHEME

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18CV72

## Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Design of RCC and Steel Structure

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any TWO full questions, choosing ONE full question from each module.  
2. Use of IS-456, IS-800, SP-16, SP(6) – steel tables are allowed.**

### Module-1

- 1 Design RCC rectangular combined footings for two columns which are 3.6 m apart carrying a load of 1000 kN and 1500 kN. Sizes of column are 400 mm × 400 mm and 600 × 600 mm. Width of footing is 1.8 m SBC of soil is 280 kN/m<sup>2</sup>. M20 concrete and Fe415 steel is used. Assume beam and slab type combined footing, sketch the details of reinforcement. (50 Marks)

OR

- 2 Design a cantilever retaining wall to retain an earth embankment with a horizontal top 3.50 m above ground level. The unit weight of back fill is 18 kN/m<sup>3</sup>. Angle of internal friction  $\phi = 30^\circ$ , SBC of soil = 180 kN/m<sup>2</sup>. Take coefficient of friction between soil and concrete = 0.55. Adopt M20 grade concrete and Fe415 grade steel. Depth of foundation = 1.0 m. (50 Marks)

### Module-2

- 3 The centre line of a roof truss is as shown in Fig.Q3. The magnitude and nature of forces under service conditions are:  
Top chord members = 120 kN compression  
Bottom chord members = 100 kN Tension  
Interior members = 60 kN Tension and 50 kN compression.  
For all the interior members use similar single angle sections. Design all the members and joints using M16 turned bolts of grade 4.6. Also design bearing plate, base plate and anchor bolts to connect the truss to an RCC column 300 mm × 300 mm of M20 grade concrete. (50 Marks)

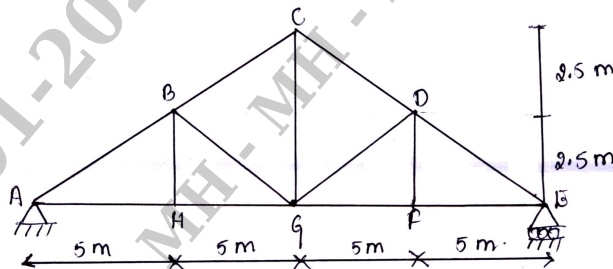


Fig.Q3

(50 Marks)

OR

- 4 Design a plate girder for an effective span 14 m. Load on the girder consist of UDL 45 kN/m in addition to two point loads each of magnitude 400 kN placed at a distance of 3m, on either side of mid span point of girder. Design Mid span cross section curtailment of flange, intermediate stiffness and end bearing stiffness. Draw sketch showing detail of longitudinal section cross section at mid span and support and plan of girder. (50 Marks)

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# CBCS SCHEME

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18CV732

## Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Air Pollution and Control

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. Assume any missing data suitably.**

### Module-1

- 1 a. Define Air Pollution. Explain the classification and sources of air pollutants. (10 Marks)
- b. Briefly explain the effects of air pollution on human health. (10 Marks)

OR

- 2 a. Briefly explain the effects of air pollution on vegetation and materials. (10 Marks)
- b. Write short notes on : (10 Marks)
  - (i) Types of Inversion
  - (ii) Photochemical Smog

### Module-2

- 3 a. With neat sketches, explain different types of plume behavior. (10 Marks)
- b. Explain the methods for measurement of meteorological variables. (10 Marks)

OR

- 4 a. Define wind rose with a neat sketch explain how a wind rose is plotted. (10 Marks)
- b. Determine the plume rise from the following data: (10 Marks)
  - (i) Physical height of stack = 180 m
  - (ii) Inside diameter of stack = 0.95m
  - (iii) Wind velocity = 2.75m/sec
  - (iv) Air temperature = 20°C
  - (v) Barometric pressure = 1000 mb
  - (vi) Stack gas velocity = 11.12m/sec
  - (vii) Stack gas temperature = 160°C

### Module-3

- 5 a. Explain in brief monitoring and analysis of following air pollutants: (10 Marks)
  - (i) PM<sub>10</sub>
  - (ii) SO<sub>x</sub>
- b. What is meant by air sampling? Explain non-isokinetic and isokinetic sampling. (10 Marks)

OR

- 6 a. Explain in brief monitoring and analysis of NO<sub>x</sub> and PM<sub>2.5</sub>. (10 Marks)
- b. Explain Gaussian dispersion model and write the assumption of the model. (10 Marks)

### Module-4

- 7 a. Write short notes on: (10 Marks)
  - (i) Settling chambers
  - (ii) Cyclone separators.
- b. With the help of neat sketch, explain the working principle of Electro Static Precipitators (ESP). (10 Marks)

OR

- 8 a. List the different types of scrubbers and explain any two with neat sketch. (10 Marks)  
b. With a neat sketch, explain the operation of Fabric filter. (10 Marks)

**Module-5**

- 9 a. Explain briefly the types of emission in automobiles and main approaches to minimize exhaust emissions. (10 Marks)  
b. Define Noise Pollution. Explain the effects and its control measures. (10 Marks)

OR

- 10 Write short notes on :  
(i) Acid Rain  
(ii) Environmental Law and Act  
(iii) Green house effect. (20 Marks)

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18CV733

## Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Pavement Materials and Construction

Time: 3 hrs.

Max. Marks: 100

**Note: Answer any FIVE full questions, choosing ONE full question from each module.**

### Module-1

- 1 a. Explain the desirable properties of aggregates to be used in pavement construction. (08 Marks)
- b. List the various tests conducted on road aggregates in order to ascertain its suitability and indicate the desirable values of the test results. (08 Marks)
- c. Briefly explain the classification of aggregates. (04 Marks)

**OR**

- 2 a. With a note on aggregate blending to meet the specified gradation. (08 Marks)
- b. What are the tests conducted on Bitumen? Mention the importance of each. (08 Marks)
- c. List and explain the chemical constituent of bituminous road binder. (04 Marks)

### Module-2

- 3 a. What are emulsions and cutbacks? How are they prepared? Mention the conditions of its use. (10 Marks)
- b. List the various lab tests to determine the adhesion of bituminous binder to an aggregate and explain any one. (10 Marks)

**OR**

- 4 a. Explain the common adhesion problem in Bituminous construction. (06 Marks)
- b. Briefly explain the mechanism of stripping of Bituminous binder. (08 Marks)
- c. List the different tests conducted on bituminous emulsions. Explain any one. (06 Marks)

### Module-3

- 5 a. List and explain the desirable properties of bituminous mix. (08 Marks)
- b. Explain the procedure of determining the optimum bitumen content for a bituminous mix by Marshall stability test. (12 Marks)

**OR**

- 6 a. Explain : (08 Marks)
  - i) Open graded mix
  - ii) Dense graded mix
  - iii) Flexibility and brittleness
- b. Explain proportioning of materials by Rothfuch's method with the help of a graph. (12 Marks)

### Module-4

- 7 a. Explain the procedure of the compacting equipments used for highway construction? Mention its uses. (10 Marks)
- b. Enumerate the steps in the preparation of subgrade. (10 Marks)

**OR**

- 8 a. Mention the equipments used for excavation in the construction of bituminous pavements. Explain any two with a neat sketch. (10 Marks)
- b. How is the adequacy of the compaction in the field evaluated? What are the quality control tests or checks in the field? (10 Marks)

**Module-5**

- 9 a. Discuss the materials required and construction steps for the construction of Bituminous concrete pavement. (10 Marks)
- b. Explain the material specification and quality control checks for wet mix macadam. (10 Marks)

**OR**

- 10 a. Mention the specification and method of cement concrete pavement construction. (10 Marks)
- b. List the quality checks on cement concrete pavement carried out both laboratory and on the field. Explain briefly. (10 Marks)

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18CV745

## Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

### Urban Transport Planning

Time: 3 hrs.

Max. Marks: 100

**Note: Answer any FIVE full questions, choosing ONE full question from each module.**

#### Module-1

- 1 a. What is Urbanisation? What are the advantages and disadvantages of Urbanisation? (10 Marks)
- b. List and explain various Urban transportation problems. (10 Marks)

**OR**

- 2 a. Briefly explain various steps involved in Urban transportation planning process. (10 Marks)
- b. Explain various types of Co-ordination in Passenger transport. (10 Marks)

#### Module-2

- 3 a. Define External Cordon Line. What factors are considered while selecting external Cordon line? (06 Marks)
- b. What are various points to be considered when dividing the Urban area into zones? (06 Marks)
- c. Briefly explain the procedure for conducting home interview surveys. What are the advantages and disadvantages of Home interview surveys? (08 Marks)

**OR**

- 4 a. Explain Registration number plate survey and Tag on vehicle survey method with their advantages and disadvantages. (10 Marks)
- b. List and explain inventory of transport facilities. (10 Marks)

#### Module-3

- 5 a. Explain Category analysis. What are the advantages and disadvantages of this method? (10 Marks)
- b. The following data is collected for a town :

| Zone                        | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-----------------------------|----|----|----|----|----|----|----|
| Population in Thousand      | 25 | 20 | 28 | 18 | 19 | 21 | 22 |
| Trips generated in Hundreds | 18 | 15 | 20 | 13 | 14 | 16 | 17 |

Develop a linear regression model for trips generated from a zone and compute the coefficient of correlation. Predict the expected trip for a zone with population 50000.

(10 Marks)

**OR**

- 6 a. List the various growth factor methods of trip distribution. Explain any one method in detail with the advantages and disadvantages. (10 Marks)

- b. Calculate the future trip distribution based on Furness method (up to two iteration) from the following data :

| O \ D        | Present trips |     |     |     | Future Trips |
|--------------|---------------|-----|-----|-----|--------------|
|              | 1             | 2   | 3   | 4   |              |
| 1            | 10            | 20  | 15  | 18  | 140          |
| 2            | 21            | 16  | 17  | 14  | 150          |
| 3            | 30            | 21  | 25  | 27  | 200          |
| 4            | 10            | 9   | 16  | 13  | 100          |
| Future Trips | 150           | 110 | 170 | 160 |              |

(10 Marks)

**Module-4**

- 7 a. Explain : i) Gravity model ii) Intervening opportunity model. (10 Marks)  
 b. Distribute 650 work trips from zone – 3 to zones 1, 2, 3, 4, and 5 from the data given below.

| Zone No | Total work trip attractions | Travel time in minutes | Travel time factor |
|---------|-----------------------------|------------------------|--------------------|
| 1       | 1080                        | 20                     | 6                  |
| 2       | 531                         | 7                      | 29                 |
| 3       | 76                          | 5                      | 45                 |
| 4       | 47                          | 10                     | 18                 |
| 5       | 82                          | 25                     | 4                  |

(10 Marks)

**OR**

- 8 a. Define Modal Split. Explain in brief factors affecting modal split. (10 Marks)  
 b. With a neat flow diagram, explain pre distribution modal split. What are the advantages and disadvantages of this method? (10 Marks)

**Module-5**

- 9 a. Define Traffic assignment. What are the applications of traffic assignment? (10 Marks)  
 b. Write short notes on :  
 i) All or nothing assignment technique.  
 ii) Capacity restraint assignment. (10 Marks)

**OR**

- 10 a. With a neat flow chart, explain the principle components of the Lowry model. (10 Marks)  
 b. Discuss the points for the selection of Land use transport modal. (10 Marks)

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## Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Environmental Protection and Management

Time: 3 hrs.

Max. Marks: 100

*Note: Answer any FIVE full questions, choosing ONE full question from each module.*

### Module-1

- 1 a. Explain unique characteristics of Environmental problems with examples. (10 Marks)
- b. Describe the classification made to environmental Impact Reduction Efforts. (10 Marks)

**OR**

- 2 a. Explain the Environmental Management Principles. (10 Marks)
- b. Explain National policies for the abatement of pollution. (10 Marks)

### Module-2

- 3 a. List and explain the environmental quality objectives. (10 Marks)
- b. What are cleaner production and clean technology? (10 Marks)

**OR**

- 4 a. Write a note on 'Closing the Loops' with industries. (10 Marks)
- b. Explain the purpose and objective of environmental performance indicators. (10 Marks)

### Module-3

- 5 a. Distinguish between EMS and EMAS, with a neat sketch and explain the components of EMS. (10 Marks)
- b. What is ISO? State the requirements and recommended approach to implementing of ISO 14000 with an organization. (10 Marks)

**OR**

- 6 a. Define environmental policy and write the note on developing environmental policy. (10 Marks)
- b. Write a note on identifying environmental aspects and impacts within an organizational set-up. (10 Marks)

### Module-4

- 7 a. Define Environmental Audit. Explain the objectives and scope of EA. (10 Marks)
- b. Explain the Environmental performance indicators and their evolution. (10 Marks)

**OR**

- 8 a. List and describe the general steps in an Environmental Auditing. (08 Marks)
- b. Write a short note on waste audit and waste minimization planning in any industry. (12 Marks)

### Module-5

- 9 a. Write in brief about air and water pollution prevention opportunities in textile industry. (10 Marks)
- b. Explain the concept of transboundary of pollutants of hazardous waste. (10 Marks)

**OR**

- 10 Explain in brief about applications of EMS applied in textile, sugar, pulp and paper, electroplating, tanning industries. (20 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.  
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.